

JUSTIN M. PARRA

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ENGINEERING PROFILE & CORE COMPETENCIES

Electrical and Computer Engineering undergraduate at **Rutgers University – New Brunswick** (graduating May 2027) specializing in embedded firmware architecture, real-time control algorithms, mixed-signal hardware design, and autonomous robotics.

Core Specializations: Bare-Metal & RTOS Firmware Architecture, Cascade PID Control, Sensor Fusion (EMA / EKF Principles), A* & Modified Flood Fill Pathfinding, 4-Layer PCB Design (Altium Designer / KiCad), and POSIX Telemetry Simulation.

Target Engineering Domains: Embedded Software Engineering, Firmware Engineering, Robotics & Controls, Microcontroller Architectures, Semiconductor/EDA Tooling, and Hardware / Digital Systems Design.

EDUCATION

Rutgers University – New Brunswick, NJ

Expected May 2027

Bachelor of Science in Electrical and Computer Engineering (ECE)

Circuits & Microelectronics: Principles of Electrical Engineering I & II (AC/DC Circuits, Transients, Phasors, Impedance, Op-Amps, Frequency Response, Resonance), Microelectronic Circuits, Semiconductor Devices, Materials Physics

Systems, Signals & Controls: Linear Systems and Signals, Control Systems Principles, Feedback & State Estimation, Sensor Fusion

Computer Engineering & Architectures: Digital Logic Design, Computer Architecture, Embedded Systems & Microcontrollers, Programming Methodology I & II (Modern C/C++, Data Structures & Algorithms)

Mathematics & Physics: Differential Equations for Engineering & Physics (MATH 244), Multivariable Calculus, Linear Algebra, Discrete Mathematics

Energy & Capstone: Sustainable Energy Systems (14:332:402), Introduction to ECE Capstone Design

COMPREHENSIVE TECHNICAL SKILLS BANK

Programming & Scripting Languages: C, C++ (C++11/17/20), Python (NumPy, SciPy, Matplotlib, PySerial), TypeScript/JavaScript, MATLAB, Java, Verilog HDL, Assembly (x86 / ARM / RISC), Bash/Shell Scripting, SQL, Spanish (Native/Bilingual)

Embedded Systems & Microcontrollers: ESP32-S3 (Dual-Core Xtensa 32-bit), STM32F405 / STM32 (ARM Cortex-M4), Arduino, FreeRTOS Multitasking, Bare-Metal C, Hardware Abstraction Layers (HAL), Direct Register-Level I/O, Timer ISRs, DMA, Bootloaders

Hardware Design & EDA Tools: Altium Designer, KiCad 8.0 (4-Layer PCB Routing, ENIG Surface Finish, Impedance Control, Coplanar Ground Shielding), EasyEDA, Digital Logic Design, Analog Signal Conditioning, Active Op-Amp Filters, RLC Transient Analysis, TVS Surge Protection, Decoupling Networks

Protocols & Communication Buses: I²C (Standard & 400 kHz Fast-Mode), SPI, UART, RS-485 Differential Bus, CAN Bus, PWM Motor Control, GPIO, ADC / DAC Conversion

Robotics & Control Theory: Dual-Loop Cascade PID Control, Discrete Velocity Control, Sensor Fusion (EMA, Complementary Filters, EKF Principles), Differential Drive Kinematics, State Estimation, Modified Flood Fill, A* Pathfinding Algorithm, Raycast Modeling

EDA, Simulation & CAD Tools: Altium Designer, KiCad, LTspice, Multisim, SOLIDWORKS (3D CAD Modeling), Qt6 Framework (MMS Simulator), ElkJS Graph Routing, 3D Printing (FDM/SLA Rapid Prototyping)

Lab Instrumentation & Verification: Mixed-Signal & Digital Storage Oscilloscopes (DSO), Logic Analyzers, Benchtop Power Supplies, Arbitrary Function Generators, Digital Multimeters (DMM), Soldering & Surface-Mount (SMD) Rework

Developer Tools & Environments: Linux (Arch Linux, Ubuntu, WSL2), Git / GitHub, CMake, Make, GDB Debugger, PlatformIO, VS Code, POSIX Inter-Process Communication (IPC)

LEADERSHIP, HONORS & AFFILIATIONS

Rutgers Micromouse (IEEE Student Branch): Co-President & Team Lead (September 2024 – Present) — Led technical onboarding workshops on embedded C++, PCB layout in Altium Designer/KiCad, and cascade PID control for 40+ engineering undergraduates; managed team operations and secured \$2,500+ in industry and departmental funding.

Rowan Regional Micromouse Competition: 1st Place Regional Champion (2025) — Designed chassis, 4-layer PCB, and firmware for the championship-winning autonomous maze robot “Jerrieee”.

IEEE Rutgers Student Chapter: Active Member & Project Lead (September 2023 – Present) — Competed in collegiate robotics design challenges, autonomous hardware hackathons, and regional IEEE technical symposiums.

ENGINEERING WORK EXPERIENCE

Rutgers Micromouse

New Brunswick, NJ

Lead Firmware & Hardware Engineer | Lead Controls Engineer | Co-President

September 2024 – Present

- Engineered real-time embedded C/C++ firmware on ESP32-S3 (dual-core Xtensa 32-bit) and STM32 microcontrollers, utilizing FreeRTOS tasks to achieve deterministic 100 Hz sensor polling and sub-millisecond motor control response.
- Architected dual-loop cascade PID feedback controllers fusing 100 Hz IMU angular velocity (Bosch BNO055 9-DOF) with lateral distance error from ST VL53L1X Time-of-Flight sensors, eliminating motor drift and achieving sub-millimeter maze centering.
- Implemented memory-efficient Modified Flood Fill exploration and A* pathfinding algorithms in modern C++ with bit-manipulated state matrices and discrete PID velocity control, reducing autonomous traversal solve time by 20% under strict embedded SRAM limits.
- Designed and routed custom 4-layer PCBs in Altium Designer and KiCad with ENIG surface finish, integrating STM32F405 / ESP32-S3, 6-DOF IMU, ToF arrays, and dual H-bridge motor drivers while optimizing trace impedance and coplanar ground shielding.
- Developed precision analog IR optical distance-sensing circuits with active ambient-light filtering and analog low-pass stages, improving obstacle detection repeatability to ± 2 mm at velocities up to 1.5 m/s.
- Characterized signal-to-noise ratios (SNR), bus rise times, and multi-slave arbitration across 400 kHz I²C and SPI lines using mixed-signal oscilloscopes and logic analyzers, eliminating communication dropouts.
- Programmed low-level register-direct I²C and SPI peripheral drivers, hardware abstraction layers (HAL), and timer ISRs to guarantee microsecond deterministic execution and glitch-free PWM motor drive.
- Built a headless POSIX simulation pipeline using Qt6 MMS under Arch Linux communicating over inter-process pipes, verifying state-machine transitions and sensor logic prior to hardware deployment.
- Developed Python mathematical modeling scripts (NumPy, SciPy, Matplotlib) to simulate motor torque-inertia curves, filter analog sensor noise, and stream real-time serial telemetry.
- Scaled active club membership by 900% (40+ student engineers across robotics hardware and software sub-teams) and secured \$2,500+ in corporate sponsorship to fund custom PCB fabrication, component procurement, and regional competitions.

LPT-KEYPAK

Tewksbury, NJ

Engineering Intern

June 2024 – August 2024

- Commissioned and validated automated high-speed packaging machinery for the Bill & Melinda Gates Foundation, boosting packaging line throughput by 30% during pilot production qualification runs.
- Diagnosed electromechanical and process failure points across automated manufacturing equipment using structured root-cause analysis (5-Whys, Pareto charts), reducing production defect rates by 15%.
- Interpreted 20+ industrial electrical schematics to troubleshoot PLCs, motor drives, and sensor relays; authored comprehensive technical documentation, Best-Known Methods (BKMs), and maintenance SOPs to standardize equipment integration and operator workflows.
- Validated sensor calibration curves, voltage tolerances, and relay timing margins using digital storage oscilloscopes and precision multimeters, ensuring strict compliance with operational safety and quality standards.
- Formulated process validation protocols and coordinated directly with field technicians and senior controls engineers to resolve intermittent electrical noise and grounding anomalies.

VEX Robotics – Rutgers IEEE

New Brunswick, NJ

Lead Electrical Engineer

September 2024 – Present

- Architected communication and distributed power delivery architectures for competition robotics, engineering RS-485 $\rightarrow$ UART $\rightarrow$ I²C transceiver networks between central VEX Brain and distributed peripheral microcontrollers.
- Designed schematics and selected components including differential transceivers, TVS transient surge diodes, decoupling filter networks, and bus termination pull-up/down resistors to protect sensitive logic from inductive spikes.
- Executed hardware-in-the-loop (HIL) stress testing and signal integrity validation under severe mechanical vibration using digital oscilloscopes to verify packet-loss rates $< 0.1\%$ and guarantee deterministic bus arbitration.
- Managed wiring harness fabrication, power budgeting across multi-motor drive subsystems, and electrical subsystem compliance under strict competition rules.

Autonomous Micromouse Robot (“Jerrieee”) — 1st Place Champion

github.com/justintime1888/Micromouse-2025

C/C++, STM32F405RG, KiCad 8.0, 4-Layer ENIG PCB, DRV8847, VL6180X ToF, Cascaded PID 2024 – Present

- Won 1st Place at Rowan Regional Competition; designed custom 4-layer ENIG PCB integrating 168 MHz ARM Cortex-M4 MCU, 5× optical ToF array over 400 kHz I²C, and dual DRV8847 motor drivers in an 84×76.5 mm chassis.
- Programmed real-time Flood-Fill maze navigation algorithm running in <850 μs per cell transition with a 1 kHz cascaded PID control loop correcting gyro heading and velocity errors up to 4.2 m/s with zero integral windup.

Weave — Automated SPICE Netlist Parser & Schematic Compiler

github.com/justintime1888/weave

TypeScript, LTspice, ElksJS Layered Graph Routing, Geometric Placement, EDA Tools 2025

- Built an automated compiler converting raw SPICE subcircuit netlists (.cir) into formatted LTspice schematics (.asc) using hierarchical graph theory and layered placement algorithms.
- Implemented geometric Manhattan routing and net-label heuristics to eliminate overlapping wire segments and generate clear, human-readable circuit schematics automatically.

Silicon Wafer Yield & Defect Clustering Simulator

github.com/justintime1888/wafer-yield-simulator

Python, NumPy, Matplotlib, Semiconductor Physics, Defect Density Modeling, Yield Curves 2025

- Modeled Poisson and negative binomial defect clustering across 300 mm silicon wafers to compute die yield curves, cost-per-good-die, and lithography edge-exclusion metrics for semiconductor manufacturing.
- Rendered interactive high-resolution wafer maps displaying spatial defect distributions and parametric yield dropouts across variable die footprints and defect densities.

Autonomous Vision Line-Following Robot

2025 – Present

Embedded C++, ESP32-S3, OV2640 Camera, FreeRTOS, PlatformIO, Closed-Loop PID Control

- Built an autonomous mobile robot using an ESP32-S3 microcontroller and OV2640 camera module, developing an onboard vision-processing pipeline in C++ to detect track curvature in real time.
- Tuned closed-loop PID control algorithms for dual DC motors, maintaining trajectory tracking stability at speeds up to 1.2 m/s with zero track departures.

POSIX Robotics Simulation & Telemetry Engine

2025 – Present

C++, Qt6, Python, NumPy, SciPy, Matplotlib, PySerial, POSIX Pipes, Arch Linux

- Built a headless simulation pipeline in Qt6/C++ benchmarking pathfinding algorithms across 100+ competition maze layouts over POSIX inter-process pipes under Arch Linux.
- Developed a real-time serial telemetry data logger and visualizer capturing motor PWM output, angular error, and ToF distance streams at 100 Hz using Python and Matplotlib to benchmark PID settling time and overshoot.

Analog Filter & Transient Signal Conditioning Analyzer

2024

LTspice, MATLAB, Signal Integrity, Circuit Modeling, AC Frequency Response, PEE II

- Simulated active 2nd-order RLC filter topologies and operational amplifier stages in LTspice, evaluating AC frequency responses, Bode phase margins, and transient damping factors.
- Designed hardware low-pass signal conditioning stages to suppress inductive switching transients from high-current DC motors, preventing ADC input distortion and verifying performance with benchtop instrumentation.

Pedestrian Traffic Control System (Hardware State Controller)

2024

Discrete Logic, 555 Timers, Decade Counters, Combinational Logic Gates, Circuit Timing Analysis, Multisim

- Designed and breadboarded a four-way traffic controller using 555 timers, decade counters, and combinational logic gates to simulate asynchronous pedestrian crossing requests.
- Analyzed clock propagation delays, race conditions, and switching thresholds with bench oscilloscopes to ensure fail-safe state transitions; authored full engineering design report.

Experimental Telemetry & Statistical Data Analysis Pipeline

2024 – 2025

Python, NumPy, SciPy, Matplotlib, Data Analysis & Visualization

- Built an automated statistical analysis pipeline in Python processing 100+ experimental test runs to evaluate transient settling metrics, overshoot variance, and parameter confidence intervals.
- Generated visual dashboards and distribution plots for experimental data interpretation, identifying process outliers and driving closed-loop system tuning.